# Sung Guk (John) Lee

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English (native) · Korean (conversational) · Japanese (conversational)

## SUMMARY

Software engineer (B.S. Computer Science, Summa Cum Laude, 2026) who built and deployed an AI English speaking app that my own students use in class. Completed the degree part-time over five years while teaching full-time.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, SQL
**Frameworks:** Next.js (App Router), React, Node.js, Flask, Jinja
**Databases:** PostgreSQL, Supabase, SQLite
**AI / APIs:** OpenAI (speech-to-text, TTS, LLM structured output, prompt design, moderation), Azure Speech
**Testing:** Vitest, Playwright end-to-end tests, pytest, CI/CD (GitHub Actions)
**Infrastructure:** Vercel, Supabase Auth & Storage, row-level security, SQL migrations, cron jobs, Git
**AI tools:** Claude Code, Codex

## PROJECTS

**Coco English** June 2026 – Present
*TypeScript · Next.js 15 · React 19 · Node.js · Supabase/PostgreSQL · OpenAI · Azure Speech · Vercel* coco-english.vercel.app

AI English speaking practice app, live in production, built for and used by my own students. It replaced a manual loop where I listened to every recording and typed out corrections. The app transcribes each answer, checks meaning and target-pattern use, and returns pronunciation feedback.

- Cut feedback latency on exact-match answers from 3.5-13s to about 700ms with a deterministic fast path that skips the LLM call when the answer matches exactly.
- Compared the production LLM with a faster, cheaper model on 22 real student answers, and used the grading errors it surfaced to tighten the production prompt.
- Vitest unit tests and Playwright end-to-end specs run on every push in a GitHub Actions CI/CD pipeline. Paid API adapters are stubbed at the boundary, so the unit suite makes no live API calls.
- Built the app with Claude Code and Codex on Next.js App Router and Node.js. It has 15 API endpoints, Vercel cron jobs, and a Supabase PostgreSQL database with 53 versioned SQL migrations.
- Built phoneme-level pronunciation scoring on Azure Speech with a per-student weakness profile, and validated the 60/40 accuracy/fluency weighting against 12 student recordings I scored by hand.

**University IT Support Ticketing System — Course project** 2025
*Python · Flask · SQLite · Jinja · pytest* *github.com/CSCI441Group5/academic-institution-ticketing-software*

- Service desk ticketing system for schools and small university departments, where commercial platforms like Zendesk and ServiceNow are too complex and expensive. Replaces the shared inboxes and spreadsheets these teams fall back on.
- Users submit requests by category; the system routes each ticket to the right department queue, with separate dashboards for student, staff, and manager roles and filtering by category, department, and status.
- Primary contributor on a five-person team, authoring the majority of the backend and the automated test suite.

## EXPERIENCE

**ESL Teacher — Sprout English** Jan. 2022 – Present
*Naju, South Korea*

- Design and deliver English curriculum for elementary and middle school students across multiple grade levels and proficiency bands.
- Build lesson materials, assessments, and pacing systems, iterating based on measured student performance.

**ESL Teacher — Ran English** 2016 – 2018, 2021

**Mandatory Military Service — Alternative Service, Republic of Korea** Jan. 2019 – Oct. 2020

## EDUCATION

**Fort Hays State University — B.S. Computer Science** May 2026

Summa Cum Laude · Dean's Honor List · Fort Hays Honor Society · Merit Scholarship Recipient